

NOTE ON A MULTIVARIATE ANALOGUE OF THE EXPONENTIAL REPRESENTATION

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Summary: A generalization of the univariate exponential representation for distribution functions to the multivariate case is considered. Marginal failure rates are defined and used for deriving explicit formulas. The bivariate case is studied in detail and simple examples are presented. The formal generalization for $n > 2$ is briefly considered.

1. Introduction

Let $T \geq 0$ be a lifetime random variable with an absolutely continuous CDF $F(t)$ and the corresponding failure rate $\lambda(t)$. A well-known and widely used in reliability, survival analysis and risk analysis exponential representation for the survival function holds in this case:

$$S(t) \equiv 1 - F(t) = \exp \left\{ - \int_0^t \lambda(u) du \right\}, \quad (1)$$

$$\lambda(t) = \frac{f(t)}{S(t)}, \quad (2)$$

where $f(t)$ denotes the corresponding probability density function. Therefore, the failure rate $\lambda(t)$ uniquely defines the CDF $F(t)$. The situation is more complex for the multivariate distributions and it will be shown in this
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note, that different multivariate failure rates should be defined in order to obtain the analogue of relation (1).

Let $T_1 \geq 0, T_2 \geq 0, \dots, T_n \geq 0$ be dependent lifetime random variables with support in $[0, \infty)$ and let $F(t_1, t_2, \dots, t_n)$ and $F_i(t_i)$, $i = 1, 2, \dots, n$ be their absolutely continuous multivariate and marginal (univariate) CDFs respectively. Denote by

$$S(t_1, t_2, \dots, t_n) = \Pr[T_1 > t_1, T_2 > t_2, \dots, T_n > t_n]$$

the joint survival probability (multivariate survival function). It is natural to define the multivariate failure rate as

$$\begin{aligned} & \lambda(t_1, \dots, t_n) \\ &= \lim_{\Delta t_1, \dots, \Delta t_n \rightarrow 0} \Pr[t_1 \leq T_1 < t_1 + \Delta t_1, \dots, t_n \leq T_n < t_n + \Delta t_n \mid T_1 > t_1, \dots, T_n > t_n] \\ &= \frac{f(t_1, \dots, t_n)}{S(t_1, \dots, t_n)} = (-1)^n \frac{\partial^n}{\partial t_1 \dots \partial t_n} \ln S(t_1, \dots, t_n). \end{aligned} \quad (3)$$

Therefore, $\lambda(t_1, \dots, t_n) dt_1 \dots dt_n + o(dt_1 \dots dt_n)$ can be interpreted as the probability of failure of all objects in intervals of time $[t_1, t_1 + dt_1), \dots, [t_n, t_n + dt_n)$, respectively, on condition that they did not fail before. For the bivariate case this type of failure rate was defined in Basu (1971).

Define the multivariate rate of the first order as

$$\begin{aligned} \lambda_i(t_1, \dots, t_n) &= \lim_{\Delta t_i \rightarrow 0} \frac{1}{\Delta t_i} \Pr[t_i \leq T_i < t_i + \Delta t_i \mid T_1 > t_1, \dots, T_n > t_n] \\ &= -\frac{\partial}{\partial t_i} \ln S(t_1, \dots, t_n) = -\frac{S^i(t_1, \dots, t_n)}{S(t_1, \dots, t_n)}; \quad i = 1, 2, \dots, n, \end{aligned} \quad (4)$$

where $S^i(t_1, \dots, t_n) = \partial S(t_1, \dots, t_n) / \partial t_i$. Thus, $\lambda_i(t_1, \dots, t_n) dt_i + o(dt_i)$ can also be interpreted as the probability of failure of the i th object in interval

$[t_i, t_i + dt_i)$ on condition that all objects did not fail in $[t_j, t_j + dt_j)$, $j = 1, 2, \dots, n$, respectively.

Similar to (3) and (4), define the failure rate of the k th order, $k = 1, 2, \dots, n$ as

$$\lambda_{i_1, \dots, i_k}(t_1, \dots, t_n) = (-1)^n \frac{S^{i_1, \dots, i_k}(t_1, \dots, t_n)}{S(t_1, \dots, t_n)}, \quad (5)$$

where $S^{i_1, \dots, i_k}(t_1, \dots, t_n) = \partial^k S(t_1, \dots, t_n) / \partial t_{i_1} \dots \partial t_{i_k}$ and the failure rate has the similar probabilistic interpretation for k objects $1 \leq i_1 < i_2 < \dots < i_k \leq n$: $\lambda_{i_1, \dots, i_k}(t_1, \dots, t_n) dt_{i_1} \dots dt_{i_k} + o(dt_{i_k} \dots dt_{i_k})$ is a probability of failure of these objects in the corresponding intervals of time on condition that all n objects did not fail in $[t_j, t_j + dt_j)$, $j = 1, 2, \dots, n$, respectively. It is clear that relation (3) is the specific case of (5) for $k = n$ and relation (4) is the specific case of (5) for $k = 1$. The reason for defining them separately is that the multivariate failure rate (3) was already considered before (e.g., in Basu (1971)) and it has an appealing interpretation. The failure rate of the first order was also considered previously in connection with the competing risks problems and was usually called "the crude failure rate" (e.g., in Yashin and Iachine (1996)). The failure rate of order $k \leq n - 1$ can also be called the marginal failure rate.

In what follows we will derive formally a general exponential representation for $S(t_1, \dots, t_n)$ and then transform it via the defined failure rates. Eventually, the important specific case of $n = 2$ will be considered in detail and some considerations on obtaining the exponential representation for $n > 2$ will be given in the Appendix.

2. The exponential representation

It is clear, that any n -variate survival function can be formally represented by the following simple and useful identity:

$$S(t_1, \dots, t_n) = S_1(t_1) \dots S_n(t_n) \exp \{A(t_1, \dots, t_n)\}, \quad (6)$$

where

$$A(t_1, \dots, t_n) = \ln \frac{S(t_1, \dots, t_n)}{S_1(t_1) \dots S_n(t_n)}. \quad (7)$$

and $S_i(t_i) = S(0, \dots, 0, t_i, 0, \dots, 0)$; $i = 1, 2, \dots, n$ are the corresponding marginal survival functions. It is clear that $A(t_1, \dots, t_n)$ can be viewed as a measure of dependence between T_i ; $i = 1, 2, \dots, n$; Lehmann (1966) and Shaked (1982). Assume that $S_i(t_i)$ and $A(t_1, \dots, t_n)$ are absolutely continuous. Then similar to the bivariate case considered by Yashin and Iachine (1999),

$$S_i(t_i) = \exp \left\{ - \int_0^{t_i} \lambda_i(u) du \right\}, \quad (8)$$

$$A(t_1, \dots, t_n) = \int_0^{t_1} \dots \int_0^{t_n} \varphi(u_1, \dots, u_n) du_1 \dots du_n,$$

where $\varphi(u_1, \dots, u_n)$ is some n -variate function. It is convenient to use the following notation:

$$H(t_1, \dots, t_n) \equiv \int_0^{t_1} \lambda_1(u) du + \dots + \int_0^{t_n} \lambda_n(u) du - \int_0^{t_1} \dots \int_0^{t_n} \varphi(u_1, \dots, u_n) du_1 \dots du_n. \quad (9)$$

Therefore, the following exponential representation can be considered as the formal generalization of the univariate case (1):

$$S(t_1, \dots, t_n) = \exp \{-H(t_1, \dots, t_n)\}. \quad (10)$$

The importance of this general representation in applications for arbitrary n still needs to be explored. Specific cases will be discussed in the next section.

It is worth mentioning, that the corresponding competing risks problem can also be formulated easily in the general form via relations (9) and (10) by

letting $t_1 = t_2 = \dots = t_n = t$:

$$\tilde{S}(t) \equiv S(t, \dots, t) = S_1(t) \dots S_n(t) \exp \{B(t)\}, \quad (11)$$

where

$$\begin{aligned} B(t) &\equiv A(t, \dots, t) = \ln \frac{\tilde{S}(t)}{S_1(t) \dots S_n(t)} \\ &= \int_0^t \dots \int_0^t \varphi(u_1, \dots, u_n) du_1 \dots du_n, \end{aligned} \quad (12)$$

$\frac{d}{dt} B(t) = \int_0^t (\varphi(u, \dots, u, t) + \varphi(u, \dots, u, t, u) + \dots + \varphi(t, u, \dots, u)) du$
and the overall (observed) failure rate is defined as

$$\lambda(t) = -\frac{d}{dt} \ln \tilde{S}(t) = \lambda_1(t) + \dots + \lambda_n(t) - \frac{d}{dt} B(t). \quad (13)$$

3. The bivariate case

The following result shows that the general representation in (9) and (10) can be written explicitly for the specific case $n = 2$ using the failure rates defined by equations (3) and (4).

Theorem. Let $F(t_1, t_2)$ be an absolutely continuous bivariate CDF. Then the following exponential representation for the survival function $S(t_1, t_2)$ holds:

$$\begin{aligned} S(t_1, t_2) &= \exp \left\{ - \int_0^{t_1} \lambda_1(u) du \right\} \exp \left\{ - \int_0^{t_2} \lambda_2(u) du \right\} \\ &\quad \exp \left\{ \int_0^{t_1} \int_0^{t_2} (\lambda(u, v) - \lambda_1(u, v) \lambda_2(u, v)) du dv \right\}, \end{aligned} \quad (14)$$

where $\lambda_i(u)$, $i = 1, 2$ are the failure rates of the marginal distributions, whereas the failure rates $\lambda(u, v)$ and $\lambda_i(u, v)$ are defined by relations (3) and (4), respectively.

Proof. From the definitions of $\lambda_i(t_1, t_2)$ in (4) and $H(t_1, t_2)$ in (9) for $n = 2$, the following useful relation can be obtained:

$$\lambda_i(t_1, t_2) = \frac{\partial}{\partial t_i} H(t_1, t_2) = \lambda_i(t_i) - \frac{\partial}{\partial t_i} A(t_1, t_2). \quad (15)$$

It is easily verified after differentiating both sides of this equation with respect to t_j (see also Finkelstein (2003) and Hougaard (2000, p. 446)) that

$$\frac{\partial^2}{\partial t_1 \partial t_2} A(t_1, t_2) = \frac{f(t_1, t_2)}{S(t_1, t_2)} - \frac{\partial}{\partial t_1} \ln S(t_1, t_2) \frac{\partial}{\partial t_2} \ln S(t_1, t_2). \quad (16)$$

This implies, from (3) and (4), that the function $\varphi(u, v)$ in (9) can be expressed as

$$\varphi(u, v) = \lambda(u, v) - \lambda_1(u, v) \lambda_2(u, v) \quad (17)$$

and formula (14) then follows immediately. ♦

If $\lambda(u, v) = \lambda_1(u, v) \lambda_2(u, v)$, then $\varphi(u, v) = 0$ so that the lifetimes T_1 and T_2 are independent. Therefore, $\lambda(u, v) / \lambda_1(u, v) \lambda_2(u, v)$ can be considered as a measure of correlation and one can formally construct bivariate distributions by assuming specific forms of this function.

Example 1. As the simplest example, consider the Gumbel bivariate exponential distribution of the form

$$S(t_1, t_2) = \exp\{-t_1 - t_2 - \delta t_1 t_2\},$$

where $0 \leq \delta \leq 1$. Then $\varphi(u, v) = -\delta$, $A(t_1, t_2) = -\delta t_1 t_2$ for this specific case (negative correlation) and

$$\lambda_i(t_1, t_2) = 1 + \delta t_j; \quad i, j = 1, 2; \quad i \neq j, \quad \lambda(t_1, t_2) = -\delta + (1 + \delta t_1)(1 + \delta t_2).$$

Let the life spans of two components in series be correlated accordingly. Then, as follows from relation (13) or directly from $S(t_1, t_2)$ for $t_1 = t_2 = t$, the failure rate of this system is $\lambda(t) = 2(1 + \delta t)$.

Remark 1. It is clear that the Marshall-Olkin bivariate distribution defined by

$$S(t_1, t_2) = \exp\{-\lambda_1 t_1 - \lambda_2 t_2 - \lambda_{12} \max(t_1, t_2)\}$$

where $\lambda_1, \lambda_2, \lambda_{12}$ are positive constants, does not possess an exponential representation of the form (14) (because it is not absolutely continuous) although the corresponding competing risks problem in accordance with relations (11)–(13) is well defined: $B(t) = \lambda_{1,2}t$.

Remark 2. The results of this section were obtained under assumption that the marginal and the bivariate distributions possess densities with respect to Lebesgue measure. The following generalization was suggested by a referee for the Marshall-Olkin bivariate distribution of Remark 1.

Define the singular measure μ on $[0, \infty)^2$ by

$$\mu([0, t_1] \times [0, t_2]) = \min\{t_1, t_2\}.$$

Then, because $t_1 + t_2 = \max\{t_1, t_2\} + \min\{t_1, t_2\}$,

$$S(t_1, t_2) = \exp\{-(\lambda_1 + \lambda_{12})t_1 - (\lambda_2 + \lambda_{12})t_2 + \lambda_{12} \min\{t_1, t_2\}\},$$

which can be written in the form of (14) using two different measures:

$$\exp\left\{-\int_0^{t_1} (\lambda_1 + \lambda_{12}) du\right\} \exp\left\{-\int_0^{t_2} (\lambda_2 + \lambda_{12}) dv\right\} \exp\left\{\int_0^{t_1} \int_0^{t_2} \lambda_{12} d\mu(u, v)\right\}.$$

This suggests that the theorem can be generalized to cases where the marginal and the bivariate distributions have densities (Radon-Nikodym derivatives) with respect to other measures. The usefulness of this extension for applications should be investigated in the future.

Example 2. This is a more meaningful example which illustrates the exponential representation (14); see Finkelstein (2003). Let the dependence structure be defined by

$$\frac{\lambda(u, v)}{\lambda_1(u, v) \lambda_2(u, v)} = 1 + \theta, \quad (18)$$

where θ is a constant and $\theta > -1$. Using equation (17), relation (12) can be written as

$$\varphi(u, v) = \theta \lambda_1(u, v) \lambda_2(u, v)$$

or, equivalently, as

$$\varphi(u, v) = \frac{\theta}{1 + \theta} \lambda(u, v). \quad (19)$$

For $\theta > 0$ (positive correlation) the corresponding bivariate survival function is defined uniquely (up to marginal distributions) and $H(t_1, t_2)$ is given by the expression

$$H(t_1, t_2) = \theta^{-1} \ln \left(\exp \left\{ \theta \int_0^{t_1} \lambda_1(u) du \right\} + \exp \left\{ \theta \int_0^{t_2} \lambda_2(u) du \right\} - 1 \right),$$

which defines the well-known Clayton copula

$$S(t_1, t_2) = \left(S(t_1)^{-\theta} + S(t_2)^{-\theta} - 1 \right)^{-\theta^{-1}}; \quad (20)$$

see Clayton and Cusick (1985) and Nelsen (1999).

4. Concluding remarks

In accordance with definition (5), the exponential representation (10) can be seen as a generalization of the univariate case (1) using the marginal ($k < n$) and 'full' ($k = n$) failure rates. The results and examples in Section 3 show that exponential representation for the bivariate case is quite useful for studying the dependence structure and constructing new types of bivariate survival distributions. The possibility of applying this approach for $n \geq 3$ still needs to be explored. Some results for $n \geq 3$ are given in the Appendix below. The formulas have a recurrence structure and the feasibility of their practical application is not yet clear.

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Appendix: Formulas for $n \geq 3$

Let $n = 3$. It follows from equation (9) that

$$\varphi(t_1, t_2, t_3) = -\frac{\partial^3 H(t_1, t_2, t_3)}{\partial t_1 \partial t_2 \partial t_3}.$$

In accordance with definitions (4), (5) and (9):

$$H(t_1, t_2, t_3) = -\ln S(t_1, t_2, t_3);$$

$$\frac{\partial H(t_1, t_2, t_3)}{\partial t_3} = -\frac{S^3(t_1, t_2, t_3)}{S(t_1, t_2, t_3)} = \lambda_3(t_1, t_2, t_3)$$

$$\begin{aligned} \frac{\partial^2 H(t_1, t_2, t_3)}{\partial t_2 \partial t_3} &= \frac{S^3(t_1, t_2, t_3)}{S(t_1, t_2, t_3)} \frac{S^2(t_1, t_2, t_3)}{S(t_1, t_2, t_3)} - \frac{S^{2,3}(t_1, t_2, t_3)}{S(t_1, t_2, t_3)} \\ &= \lambda_3(t_1, t_2, t_3) \lambda_2(t_1, t_2, t_3) - \lambda_{2,3}(t_1, t_2, t_3). \end{aligned} \quad (\text{A-1})$$

Finally differentiating (A-1) with respect to t_1 :

$$\begin{aligned} \frac{\partial^3 H(t_1, t_2, t_3)}{\partial t_1 \partial t_2 \partial t_3} &= \lambda_{1,2,3}(t_1, t_2, t_3) - (\lambda(t_1, t_2, t_3) \lambda_{2,3}(t_1, t_2, t_3) \\ &\quad + \lambda_2(t_1, t_2, t_3) \lambda_{1,3}(t_1, t_2, t_3) + \lambda_3(t_1, t_2, t_3) \lambda_{1,2}(t_1, t_2, t_3)) \\ &\quad + 2\lambda_1(t_1, t_2, t_3) \lambda_2(t_1, t_2, t_3) \lambda_3(t_1, t_2, t_3) = -\varphi(t_1, t_2, t_3) \end{aligned} \quad (\text{A-2})$$

and substituting equation (A-2) in relations (9) and (10), we arrive at the corresponding exponential representation for $n = 3$. The 'structure' of recurrent differentiation in (A-1) is quite clear and can be used for deriving corresponding formulas for $n > 3$. For instance, for $n = 4$ differentiation

with respect to t_4, t_3, t_2 in accordance with relation (A-2) gives

$$\begin{aligned} \frac{\partial^3 H(t_1, t_2, t_3, t_4)}{\partial t_2 \partial t_3 \partial t_4} &= \lambda_{2,3,4}(t_1, t_2, t_3, t_4) - (\lambda_2(t_1, t_2, t_3, t_4) \lambda_{3,4} \\ &\quad (t_1, t_2, t_3, t_4) + \lambda_3(t_1, t_2, t_3, t_4) \lambda_{2,4}(t_1, t_2, t_3, t_4) \\ &\quad + \lambda_4(t_1, t_2, t_3, t_4) \lambda_{2,3}(t_1, t_2, t_3, t_4)) \\ &\quad + 2\lambda_2(t_1, t_2, t_3, t_4) \lambda_3(t_1, t_2, t_3, t_4) \lambda_4(t_1, t_2, t_3, t_4). \end{aligned}$$

The last differentiation with respect to t_1 results in

$$\begin{aligned} -\varphi(t_1, t_2, t_3, t_4) &= \frac{\partial^3 H(t_1, t_2, t_3, t_4)}{\partial t_1 \partial t_2 \partial t_3 \partial t_4} = -\lambda_{1,2,3,4} \\ &\quad + (\lambda_{1,2,3} \lambda_4 + \lambda_{1,3,4} \lambda_2 + \lambda_{2,3,4} \lambda_1 + \lambda_{1,2,4} \lambda_3) \\ &\quad + (\lambda_{1,2} \lambda_{3,4} + \lambda_{1,3} \lambda_{2,4} + \lambda_{1,4} \lambda_{2,3}) \\ &\quad - 2(\lambda_{1,2} \lambda_3 \lambda_4 + \lambda_{1,3} \lambda_2 \lambda_4 + \lambda_{1,4} \lambda_2 \lambda_3 \\ &\quad + \lambda_{2,3} \lambda_1 \lambda_4 + \lambda_{2,4} \lambda_1 \lambda_3 \\ &\quad + \lambda_{3,4} \lambda_1 \lambda_2) + 6\lambda_1 \lambda_2 \lambda_3 \lambda_4, \end{aligned}$$

where the corresponding arguments in the failure rates are omitted for brevity.